

Sea Scout Predicted-Log Navigation Rally — Rules

1. General

1.1 The rally is a predicted-log and dead-reckoning navigation passage. Each ship predicts the time to run each leg of the course, then runs the course without access to a timepiece. The lowest percentage of error wins. It is not a speed contest.

1.2 Ships shall plan for the passage duration specified in the event instructions at their safe declared cruising setting.

1.3 **No ship is disqualified in this event.** Where these rules would otherwise disqualify a ship, the run is recorded as **training-only**: the ship completes the course, receives a full debrief, and is not eligible for awards.

1.4 Safety action always takes precedence over scoring.

2. The Course

2.1 The event course instructions shall define the start and finish lines and the required direction of crossing each line.

2.2 The event course instructions shall identify the control points. **Bridges are optional control points** that may be included in the event course; when included, the instructions shall specify the span to pass **directly under**.

2.3 The event course instructions shall specify the permitted direction or directions and the order of control points for each direction.

2.4 Deviation from the direct track between control points shall be the minimum required for safety and good seamanship.

3. Event Window and Start Time

3.1 The event instructions shall specify the event window, including the earliest permitted start time and the finish deadline. No ship may cross the start line before the specified opening time.

3.2 Each crew selects its own start time within the window. The start time is a navigational planning decision — crews are expected to choose it to work the tidal streams over the course as a whole.

3.3 Every ship should be across the finish line by the event's finish deadline. Late finishes are governed by Rule 7.3.

4. The Predicted Log

4.1 Before starting, each ship shall submit a paper Predicted Log to the Contest Committee showing:

- planned route and **declared direction**;
- **declared start time**;
- declared base RPM or repeatable propulsion setting;
- estimated through-water speed;
- estimated current for each leg;
- predicted cumulative elapsed time at every control point;
- predicted elapsed time at the finish.

4.2 **Lock rule.** Once the Predicted Log has been handed to the Contest Committee, the **declared direction and declared start time may not be changed** — not before the start, and not underway. Predictions may not be altered. Before the log is handed in, a ship may change either freely.

4.3 A ship that runs the course in a direction other than the one declared on its submitted Predicted Log is scored **training-only**.

5. Divisions

Division	Youth navigation aids	Purpose
Dead Reckoning / Traditional	Paper chart, compass, tachometer, through-water log if fitted, depth sounder, basic calculator, visual bearings and ranges. No GPS position, moving map, SOG/COG, route guidance, electronic time, or AIS target display.	Primary learning division
Open Electronics	All normal vessel navigation equipment, including GPS/chartplotter, radar, AIS, autopilot and electronic ranging. No visible time-of-day or elapsed-time display to youth.	Compare modern navigation against DR planning
Training	Same as Dead Reckoning, but the observer may coach and GPS may be exposed as required for safety.	Appropriate for first-time crews

6. Rules Underway

6.1 Propulsion setting

Each ship shall declare a base RPM or repeatable propulsion setting. During the scored passage the ship may increase RPM by no more than **10%** above the declared setting. The operator may reduce RPM, change speed, or maneuver as necessary for safe operation, traffic, wakes, weather, navigation, mechanical concerns, or the directions of event officials. The observer shall record every material speed change with its reason.

6.2 Observer and safety

Each ship shall carry a designated adult observer. The observer records official times, verifies the required control points, documents material speed or course deviations, and retains the ship's actual GPS track for post-event review.

The qualified adult operator may use any navigation or communication equipment, vary speed, or depart from the course whenever necessary for safety.

6.3 Communications

- Monitor **VHF Channel 16** at all times unless actively communicating on another channel.
 - Monitor the designated event working channel when operationally appropriate.
 - Use the event working channel only for check-in, safety, withdrawal, schedule changes, and official directions.
 - Ships shall **not** exchange navigation, timing, mark-location, current, or course-performance information with competing ships.
-

7. Timing and Control Points

7.1 Elapsed time

Elapsed time is measured **line to line** — from the ship's actual crossing of the start line to its actual crossing of the finish line.

7.2 Start-time deviation penalty

If the actual start-line crossing differs from the declared start time by more than **five minutes**, early or late, the amount by which the difference exceeds five minutes is added to the ship's total accumulated time error.

Actual start vs. declared	Penalty added
3 min early or late	none
5 min early or late	none
7 min late	2 min
12 min early	7 min

7.3 Late-finish penalty

A ship finishing after the event's finish deadline has **twice its overtime** added to its total accumulated time error. The penalty is capped at **30 minutes**, reached at 15 minutes of overtime.

Finish relative to the event's deadline	Penalty added
On time or earlier	none
5 min late	10 min
10 min late	20 min
15 min late	30 min (maximum)
More than 15 min late	Training-only

7.4 Safety relief from the late-finish penalty

If a ship finishes late for safety or safety-related reasons — a safety deviation, an order of an authority, a mechanical failure, or rendering aid — **and the adult observer has logged the start and end times of that event**, the run may still be scored as a **qualifying race**.

The scorer deducts the logged duration of the event from the ship's finish time. If the resulting projected finish time falls within the event window and inside the limits of Rule 7.3, the run is scored normally and the late-finish penalty is waived to the extent covered by the logged time.

The observer's log is the sole basis for this relief. **Time that was not logged cannot be deducted.** Where the projected finish time still falls outside the limits, the event chair may waive or reduce the remaining penalty, or record the run as training-only.

8. Recording Times at Control Points

8.1 Who holds the clock. The adult observer holds the single official timepiece for the ship.

8.2 Clock setting. Before the start, the observer and the ship's skipper together verify the official timepiece against the committee's official time. Both initial the log. The time source used is recorded.

8.3 The start. The ship reports to the start line in time for its declared start time and advises the committee station on the event working channel. The observer records the **actual** time the ship crosses the designated start line in the direction specified by the event course instructions, in hours, minutes and seconds, and enters it on the log alongside the declared start time. Elapsed time for scoring runs from this actual crossing. The observer also records the difference between the declared and actual start times for the scorer.

8.4 Securing timepieces. Immediately after the start crossing, the observer collects every portable timepiece aboard and takes custody of it. The ship covers or muffles all fixed clocks. All time- and speed-indicating displays — clocks, engine hour meters, speedometers, GPS time, ETA and time-to-go readouts — are covered with opaque tape or switched off.

8.5 Calling the mark. The youth navigator calls "**Mark**" aloud at the instant the control point is abeam — that is, at the moment the control point lies at 90° to the ship's heading on the track from the previous control point. When a bridge is designated as a control point, the mark is called at the instant the ship passes **directly under the span specified in the event course instructions**. At the start and finish it is called as the ship crosses the line. All calls are made by **visual observation**, not by chartplotter or GPS position.

8.6 What the observer writes. On the call, the observer immediately records on the Contest Log, in ink:

- the control point identifier;
- the time in **hours, minutes and seconds**;
- the side on which the control point was passed (port or starboard);
- the estimated distance off;
- the name of the person who called it.

Nothing is recorded from memory after the fact.

8.7 Silence rule. The observer does not read the time aloud, does not repeat it back, and does not disclose any actual or elapsed time to any member of the crew until after the finish mark has been called. If the crew asks, the answer is "recorded."

8.8 Scrutinizer (optional). The skipper may designate one crew member who has had no access to the predicted times as Scrutinizer. The Scrutinizer may look at the observer's official timepiece to verify that mark times, passing side and distance are being recorded correctly, but holds no timepiece and may not communicate any time to anyone aboard. If the Scrutinizer and observer disagree on any entry, **both versions are written on the back of the log** and the committee resolves it.

8.9 Missed or doubtful control points. If a control point is not found, not identified, or the call is doubtful, the observer writes that plainly on the log with the time and the circumstances, and the ship continues. The crew does **not** turn back or circle.

8.10 Stops and interruptions. If the ship stops or is interrupted for safety, by order of an authority, for mechanical failure, or to render aid, the observer records the **time out and the time in** and the reason. The crew is not told the elapsed duration. These logged times are the basis for relief under Rule 7.4.

8.11 Safety deviations. Any material speed change or course deviation is recorded by the observer with its time and its reason.

8.12 At the finish. The observer records the **actual** time of the finish-line crossing in hours, minutes and seconds; this ends the elapsed time. If the crossing is after the event's finish deadline, the observer records the overtime and any safety, authority, mechanical or rendering-aid circumstance for the event chair. The observer then returns all timepieces to the crew, keeps the official timepiece, and signs the log.

8.13 Submitting the log. The observer and the youth navigator **jointly** deliver the completed log to the scorer. The observer certifies the official times and exceptions; the youth navigator certifies the DR record and explains any difference between the plan and the observed passage. Both sign.

8.14 GPS track. The observer retains the ship's actual recorded GPS track and hands it in with the log for the debrief. It is used for teaching, not for timing.

9. Scoring

9.1 At each control point and the finish, the observer records the actual elapsed time without disclosing it to the youth crew.

9.2 The error at each control point is:

$$\text{error \%} = \frac{|\text{actual cumulative elapsed time} - \text{predicted cumulative elapsed time}|}{\text{predicted total elapsed time}} \times 100$$

9.3 The overall score is the **sum** of the individual control-point errors. **Lowest score wins.**

9.4 Both timing penalties (Rules 7.2 and 7.3) are added to the ship's **total accumulated absolute time error**, in seconds, before the error percentage is computed. Neither is applied to a single leg.

9.5 **Missed control point.** Where a control point is not found or not called, the two legs sharing that point are scored as a single leg, with a penalty set by the committee in the contest instructions.

9.6 **Training-only runs.** The event chair may record a run as training-only where safety equipment has been used for navigational guidance, where a material safety deviation has occurred, or where a ship ran a direction other than the one declared. A training-only run is completed and debriefed but is not eligible for awards.

9.7 **Logs.** An incomplete or illegible log triggers a debrief and, if necessary, removal from awards — not punitive points.

9.8 **Awards:**

- Overall lowest error
- Best individual control-point accuracy
- Best Current Estimate
- Best DR Plot
- Best Recovery / Seamanship

10. Debrief

10.1 Each crew shall conduct a post-passage review of its original plan, DR track, current predictions, actual timing record, and retained GPS track.

10.2 The crew shall identify at least **one current-related effect** and **one improvement** for a future passage.